

Nuclei in the hypothalamus

Most of the nuclei have distinct functions. They secrete hormones that act on the pituitary gland, stimulating it to produce hormones that will help achieve homeostasis in the required part of the body.

Inhibits eating and reduces food intake

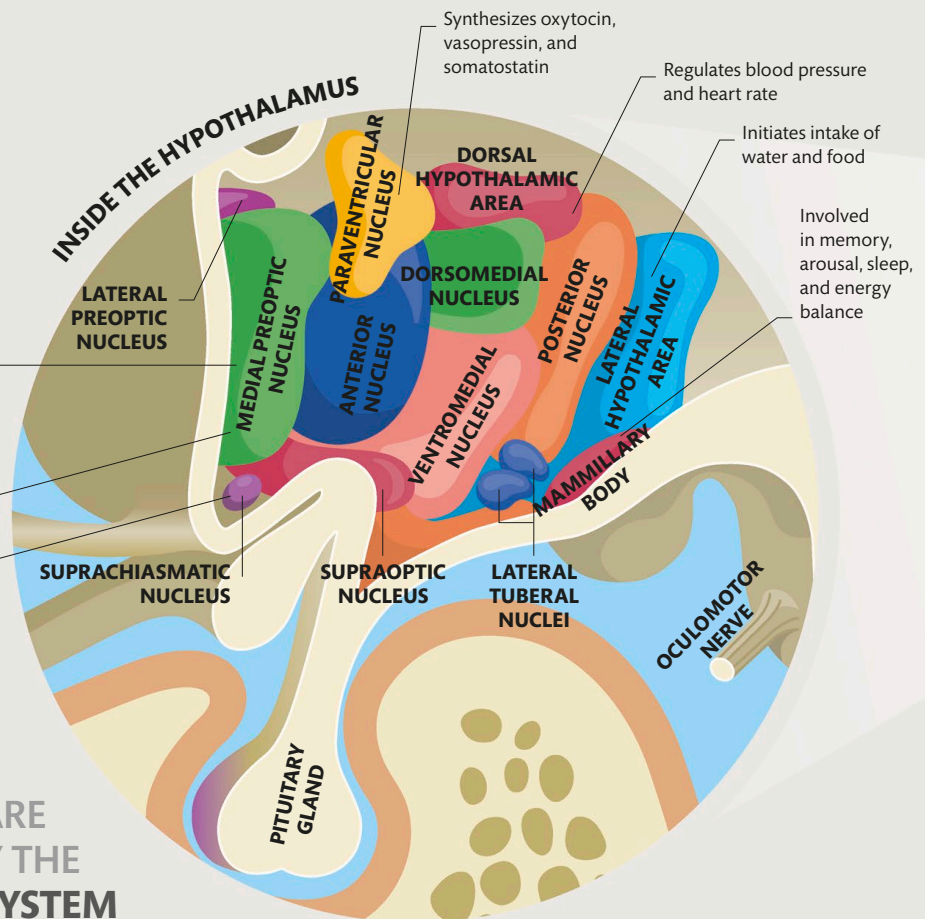
Controls thermoregulation

Body's "clock"—controls circadian rhythms



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**HORMONES ARE
PRODUCED BY THE
ENDOCRINE SYSTEM**



Neuroendocrine System

Maintaining homeostasis (see p.90) requires the brain and body to communicate. This is achieved using chemical messengers called hormones.

The hypothalamus

At the center of the brain's homeostasis system is the hypothalamus (see p.34). It contains clusters of neurons, called nuclei, that perform specific functions and has connections to the autonomic nervous system (see p.13), through which it sends messages to control heart rate, digestion, and breathing. When the hypothalamus receives a signal from the nervous system, it secretes neurohormones, which in turn stimulate the pituitary gland to secrete hormones. These affect organs all over the body and prompt them to increase or suppress their own hormone production.

OUT OF BALANCE

When homeostasis is disrupted, it can lead to disease, as well as to our cells malfunctioning. The body tries to correct the problem but may make it worse, depending on what is influencing the imbalance. Genetics, lifestyle, and toxins can all impact homeostasis.

